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- j represent data, in a graphical form, for elements 1 to 36 and use this to explain the meaning of the term 'periodic property'
 - k explain trends in the following properties of the element from periods 2 and 3 of the periodic table:
 - i melting temperature of the elements based on given data using the structure and the bonding between the atoms or molecules of the element
 - ii ionization energy based on given data or recall of the shape of the plots of ionization energy versus atomic number using ideas of electronic structure and the way that electron energy levels vary across the period.

1.6 Bonding

Students will be assessed on their ability to:

- 1 Ionic bonding
 - a recall and interpret evidence for the existence of ions, limited to physical properties of ionic compounds, electron density maps and the migration of ions, eg electrolysis of aqueous copper chromate(VI)
 - b describe the formation of ions in terms of electron loss or gain
 - c draw electron configuration diagrams of cations and anions using dots or crosses to represent electrons
 - d describe ionic crystals as giant lattices of ions
 - e describe ionic bonding as the result of strong net electrostatic attraction between ions
 - f recall trends in ionic radii down a group and for a set of isoelectronic ions, eg N^{3-} to Al^{3+}
 - g recall the stages involved in the formation of a solid ionic crystal from its elements and that this leads to a measure value for the lattice energy (students will not be expected to draw the full Born-Haber cycles)

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- h test the ionic model for ionic bonding of a particular compound by comparison of lattice energies obtained from the experimental values used in Born-Haber cycles, with provided values calculated from electrostatic theory
 - i explain the meaning of the term polarization as applied to ions
 - j demonstrate an understanding that the polarizing power of a cation depends on its radius and charge, and the polarizability of an anion depends on its size
 - k demonstrate an understanding that polarization of anions by cations leads to some covalency in an ionic bond, based on evidence from the Born-Haber cycle
 - l use values calculated for standard heats of formation based on Born-Haber cycles to explain why particular ionic compounds exist, eg the relative stability of MgCl_2 over MgCl or MgCl_3 and NaCl over NaCl_2 .

2 Covalent bonding

- a demonstrate an understanding that covalent bonding is strong and arises from the electrostatic attraction between the nucleus and the electrons which are between the nuclei, based on the evidence:
 - i the physical properties of giant atomic structures
 - ii electron density maps for simple molecules
- b draw electron configuration diagrams for simple covalently bonded molecules, including those with multiple bonds and dative covalent bonds, using dots or crosses to represent electrons.

3 Metallic bonding

- a demonstrate an understanding that metals consist of giant lattices of metal ions in a sea of delocalised electrons
- b describe metallic bonding as the strong attraction between metal ions and the sea of delocalised electrons
- c use the models in 1.6.3a and 1.6.3b to interpret simple properties of metals, eg conductivity and melting temperatures.